



Nan Jiang

Beijing University of Posts and Telecommunications(BUPT), China

Supervisor: Shangguang Wang (**Associate Dean**, School of CS)

Lab: State Key Laboratory of Networking and Switching Technology

Major: Computer Science and Technology (**Double First Class**)

+86-15901380807

✉ jn_work@outlook.com

✉ jn_bupt@bupt.edu.cn

EDUCATION

- **Beijing University of Posts and Telecommunications(BUPT), China** 9/2022-6/2025
Graduate Student - Computer Science and Technology CGPA: 90.52/100
- **Beijing University of Posts and Telecommunications(BUPT), China** 9/2018-6/2022
Bachelor's Degree - Computer Science and Technology CGPA: 3.49/4

PUBLICATIONS

- **Nan Jiang, Haitao Yuan, Jianing Si, Minxiao Chen and Shangguang Wang, "Towards Effective Next POI Prediction: Spatial and Semantic Augmentation with Remote Sensing Data" *accepted by ICDE 2024 (40th IEEE International Conference on Data Engineering)*. <https://ieeexplore.ieee.org/document/10597697>**
- **Minxiao Chen, Haitao Yuan, Nan Jiang, Zhifeng Bao, and Shangguang Wang, "Urban Traffic Accident Risk Prediction Revisited: Regionality, Proximity, Similarity and Sparsity" *accepted by CIKM 2024*. <https://arxiv.org/pdf/2407.19668>**
- **Zhengyang Zhao, Haitao Yuan, Nan Jiang, Minxiao Chen, and Shangguang Wang, "STMGF: An Effective Spatial-Temporal Multi-Granularity Framework for Traffic Forecasting" *accepted by DASFAA 2024*. <https://arxiv.org/abs/2404.05774>**

RESEARCH

- **Prompt information filtering RAG model for recommendation task on LLMs** 03/2024-present
Under the supervision of Dr. Haitao Yuan, research fellow of NTU. On-going research NTU, Singapore
 - Utilize the concept of RAG to extract information from the spatio-temporal knowledge graph, and train an external prompt generator to equip the LLM with spatio-temporal prediction capabilities.
 - Utilize GNNs to extract high-value nodes from the knowledge graph to describe user trajectory preference patterns, feeding it as prompts for the large model to predict the user's next visit.
 - Apply the Reinforcement Learning DDPG method to optimize the effectiveness of the preference selection using online human feedback.
- **Exploration on Lorenz Space for Improved Trajectory Similarity Matching** 12/2023-07/2024
Participated research. Paper submitted to Sigmod 2025. BUPT, Beijing, China
 - Introduced the Lorenz space into trajectory similarity matching tasks to overcome limitations arising from the triangle inequality constraints in traditional Euclidean geometric spaces, with partial effects validated in experiments.
 - Participated in the derivation of formulas and experimental testing work.
- **Spatio-Temporal Series Recommendation Research for Urban Data Mining** 6/2022-12/2023
Under the supervision of Dr. Haitao Yuan, research fellow of NTU. Paper accepted by ICDE 2024. NTU, Singapore
 - Introduced a spatial-semantic separated two-step prediction model for the next point-of-interests(POI) recommendation problem that can incorporate multi-modal data from Satellite-Earth scenario.
 - Proposed an innovative method that incorporates Remote Sensing Imagery to augment environmental aspects to urban region representation.
 - Developed a novel spatial graph construction approach that integrates quad-tree with POI distribution, providing an innovative strategy for user historical check-in knowledge learning.
- **Representation Learning For Trajectory Similarity Measurement** 10/2022-12/2023
Participated research. Paper submitted to ICDE 2025. BUPT, Beijing, China
 - Designed a method for aligning trajectory space and similarity space with consideration for efficiency.
 - Implemented a knowledge distillation model that distills the representation of dual-trajectory similarity into a single-trajectory embedding generation model, synthesizing the efficiency and accuracy advantages of both models.
 - Mainly Participated in the experimental testing and paper composition.
- **Versatile Framework for Urban Traffic Accident Risk** 9/2022-11/2023
Participated research. Paper accepted by CIKM 2024. BUPT, Beijing, China
 - Utilised remote sensing images to reveal regional backgrounds which indicate similar traffic accident patterns.

- Create a spatio-temporal griding method along with multi-level semantic similarity graph to inherently consider both spatial and semantic aspects on multi-granularity
- Mainly Participated in the idea construction and paper composition.

•Research on Large Scale Graph Neural Network Pre-training

2/2022-6/2022

Under the supervision of the **Prof. Chuan Shi, BUPT. Bachelor's Thesis**

BUPT, Beijing, China

- Studied the implementation of graph neural networks, conducted research on their advancements, and compared various graph pre-training methods.
- Developed a pre-training task based on min-cut algorithms and masked mechanism to address the issue of insufficient labeling in graph data.
- Designed a fast minimum cut algorithm based on the Stoer-Wagner algorithm, utilizing matrix operations to enhance the efficiency of data preprocessing for graph minimum cut computation.
- Implemented a pre-training method with node-level self-supervised training and graph-level supervised training two-step architecture.

INTERNSHIP EXPERIENCE

•VMware, Inc

5/2022-8/2022

Software Engineer (Intern)

Beijing, China

- Developed a solution to resolve audio stuttering issues of the virtual machine front-end program on macOS, greatly enhancing the user audio playback experience.
- Acquired proficiency in the company's front-end development framework(Angular) and implemented the use of the Opus audio decoder to standardize front-end audio decoding strategies
- Participated in the development team for the virtual machine front-end program and assisted in resolving a critical issue in the client application.

ACTIVITIES AND RESPONSIBILITY

•Graduation Project Guidance

12/2022-6/2023

- Assisted senior undergraduate students in researching graduation project. Topics related to Spatial-Temporal big data of Satellite-Earth scenario.
- Facilitated the design of deep learning models and provided guidance in conducting experiments and writing research papers.

•Graduate Entrance Examination Experience Lecturer

4/2023

- Invited by faculty members of the graduate group to share my experience with undergraduate students regarding graduate examinations.
- Provided explanations of study materials and shared relevant resources to undergraduate students, while also guiding them in developing effective revision plans.

•School Skateboard Club Vice President

5/2020-4/2021

- Organized fellowship, garden party and other community activities.
- Managed funds and planned community development.

AWARDS AND HONORS

•"Internet+" College Student Innovation and Entrepreneurship Competition Award

8/2023

Won the first prize in the Higher Education Main Track of the Beijing Division Semi-finals of the 9th China International 'Internet+' College Students Innovation and Entrepreneurship Competition. Also won the first Prize in the Beijing University of Posts and Telecommunications Competition Area.

•BUPT First-Class Scholarship

11/2023

Awarded for outstanding academic performance and moral acts during the academic year of post-graduate study.

•Huawei Developer Competition Award

11/2022

Participated in the code challenge for space development, led a team in designing and implementing a cloud-native satellite computing platform visualization monitoring project, and won the third prize.

•NXP Cup Intelligent Car Competition Award

6/2019

Participated in Outdoor Electromagnetic Group's competition and received awards in the university competition and won the third prize in the North China regional competition.

TECHNICAL SKILLS

Language: English(CET 6, IELTS 7.5)

Computer Skills: Python, C, C++, Javascript, Linux, Sql, Java, MATLAB

Develop Tools: Pycharm, Jupyter notebook, Visual Studio, VScode, MySql

Software Skills: Visio, Latex, Photoshop, Premiere Pro